

PRE-MEDICAL : ENTHUSIAST COURSE (PHASE - I)

This Booklet contains 20 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet and fill in the particulars on Side-1 and Side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours duration and this Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
3. Use Blue/Black Ball Point Pen only for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is *not* permissible on the Answer Sheet.

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) :_____

Candidate's Signature :_____ Invigilator's Signature :_____

Your Target is to secure Good Rank in Pre-Medical 2020

1. A point charge $q = 100 \mu\text{C}$ is located in the x-y plane at the point of position vector $\vec{r}_0 = 2\hat{i} + 3\hat{j} + \hat{k}$. What is the electric field vector at the point of position vector $\vec{r} = 8\hat{i} - 5\hat{j} + \hat{k}$?

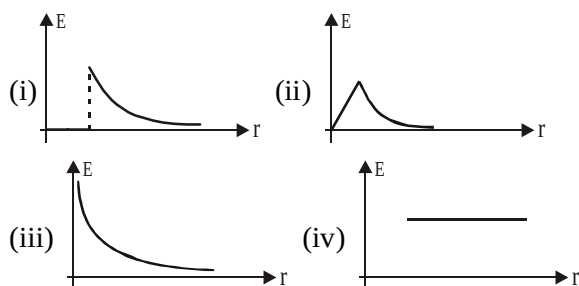
(1) 1 kV/m (2) 1 kV/cm (3) 9 kV/m (4) 9 kV/cm

2. There are two point charges q_1 and q_2 lying on a circle of unit radius. Electric field intensity at the center of circle due to these charges is $\vec{E}$. Find the position vector of the center with respect to q_2 if the position vector of the center with respect to q_1 is $\vec{r}_1$.

(1) $\frac{\vec{E} + kq_1\vec{r}_1}{kq_2}$ (2) $-\left(\frac{\vec{E} + kq_1\vec{r}_1}{kq_2}\right)$
 (3) $\frac{kq_1\vec{r}_1 - \vec{E}}{kq_2}$ (4) $\frac{\vec{E} - kq_1\vec{r}_1}{kq_2}$

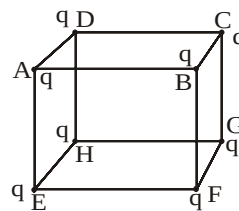
3. The value of the electric field that will hold a negatively charged particle stationary i.e. keep it balanced with respect to the gravitational force (the particle has a mass of 1 kg and a charge of 6 nC) is most nearest to
 (1) 1700 GN/C (2) 170 GN/C
 (3) 17 GN/C (4) 1.7 GN/C

4. Following are graphs of electric field variation with distance (r) for four different charge configurations. Choose the correct correspondence.



- (I) infinitely large charged sheet
 (II) infinite line charge
 (III) uniformly charged spherical shell
 (IV) uniformly charged solid sphere
 (1) (i)(II), (ii)(III), (iii)(IV), (iv)(I)
 (2) (i)(III), (ii)(IV), (iii)(I), (iv)(II)
 (3) (i)(III), (ii)(IV), (iii)(II), (iv)(I)
 (4) (i)(IV), (ii)(III), (iii)(II), (iv)(I)

5. Eight point charges having magnitude q are fixed at vertices of a cube. The electric flux through square surface ABCD of the cube is



(1) $\frac{q}{24 \epsilon_0}$ (2) $\frac{q}{12 \epsilon_0}$
 (3) $\frac{q}{6 \epsilon_0}$ (4) $\frac{q}{3 \epsilon_0}$

6. A point charge exist at origin. If potential of points A & B separated by radial distance 2 m having potentials 2 V & 1 V respectively, what is the magnitude of $\vec{E}$ at point A. (A & B lies in same quadrant).

(1) 1 V/m (2) 2 V/m
 (3) 3 V/m (4) 4 V/m

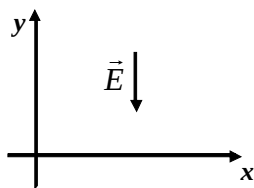
7. An electric field $(-30\hat{i} + 20\hat{j}) \text{ Vm}^{-1}$ exists in the space. If potential at the origin is zero then potential at (5m, 3m) is given by

(1) 60 V (2) 90 V
 (3) 150 V (4) 240 V

8. The potential on the N^{th} shell due to N concentric shells having charges $Q, 2Q, 3Q, \dots, NQ$ and radii $a, 2a, 3a, \dots, Na$ respectively is-

(1) $\frac{Q(N+1)}{8\pi \epsilon_0 a}$ (2) $\frac{QN(N+1)}{8\pi \epsilon_0 a}$
 (3) $\frac{Q}{2\pi \epsilon_0 a}$ (4) $\frac{Q(N+1)}{2\pi \epsilon_0 a}$

9. A particle of mass m and charge $-q$ is projected from the origin with a horizontal speed v into an electric field of intensity E directed downward. Choose the wrong statement. (Consider $+y$ axis in upward direction and x -axis in horizontal direction. Also neglect the effect of gravity)



- (1) The kinetic energy after a vertical displacement y is qEy .
 (2) The horizontal and vertical components of

acceleration are $a_x = 0$, $a_y = \frac{qE}{m}$

- (3) The equation of trajectory is $y = \frac{1}{2} \left(\frac{qEx^2}{mv^2} \right)$
 (4) The horizontal and vertical displacements

x and y after a time t are $x = vt$ and $y = \frac{1}{2} a_y t^2$

10. Electric field E in a system is given as $\vec{E} = x^2 \hat{i}$, where x is in meter and E is in (N/C). A dipole $\vec{p} = 2\hat{j}(C-m)$ is placed in the field. Then :-

- (1) Potential energy of the dipole at $x = 2$ m is 8 J
 (2) Potential energy of the dipole at $x = 2$ m is zero
 (3) Potential energy of the dipole at $x = 1$ m is 4 J
 (4) Potential energy of the dipole at $x = 2$ m is -8 J

11. A small rigid object carries positive and negative charge of magnitude 4 coulomb each. It is oriented so that the positive charge has coordinates $(-1.2\text{mm}, 1.1\text{mm})$ and negative charge has coordinates $(1.4\text{mm}, -1.3\text{mm})$. The object is kept in an electric field of $(2500\hat{i} - 5000\hat{j})$ N/C. Find the magnitude of torque acting on the dipole.

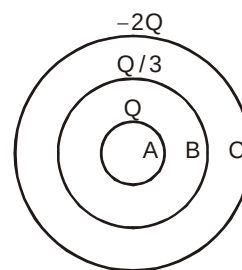
- (1) 1.4 N-m (2) 14 N-m
 (3) 2.8 N-m (4) 28 N-m

12. An electric dipole (dipole moment p) is placed at a radial distance $r \gg a$ (where a is dipole length) from an infinite line of charge having linear charge density $+\lambda$. Dipole moment vector is aligned along radial vector $\vec{r}$ force experienced by dipole is :-

(1) $\frac{\lambda p}{2\pi\epsilon_0 r^2}$, attractive (2) $\frac{\lambda p}{2\pi\epsilon_0 r^3}$, attractive

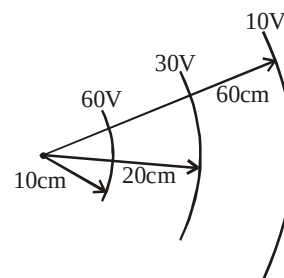
(3) $\frac{\lambda p}{2\pi\epsilon_0 r^2}$, repulsive (4) $\frac{\lambda p}{2\pi\epsilon_0 r^3}$, repulsive

13. Three conducting concentric spherical shells of radius R , $2R$ and $3R$ have charges Q , $\frac{Q}{3}$ and $-2Q$ respectively. The intermediate shell is now grounded. Find the charge flow into the earth.



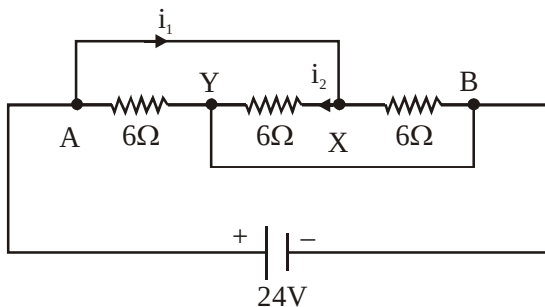
- (1) $\frac{Q}{3}$ (2) $\frac{2Q}{3}$ (3) Q (4) 0

14. Referring to the spherical equipotential lines, Express electric field as function of r . If your answer is $\frac{a}{r^2}$ then give the value of a . (Here r & a in SI unit)

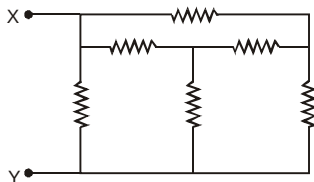


- (1) -3 (2) -6 (3) +6 (4) +9

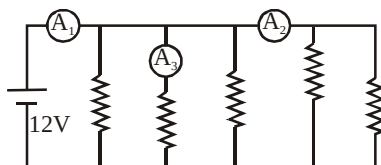
15. When a current of 4 A flows within a battery from its positive to negative terminal, the potential difference across the battery is 12 volts. The potential difference across the battery is 9 volts when a current of 2 A flows within it from its negative to its positive terminal. The internal resistance and the e.m.f. of the battery are :-
 (1) 0.1 Ω , 4V (2) 0.2 Ω , 5V
 (3) 0.5 Ω , 10V (4) 0.7 Ω , 10V
16. In the figure shown, the current in the wires which join the points A and X is i_1 and the points X and Y is i_2 . Then i_1 & i_2 are respectively ?



- (1) $i_1 = 12$ A, $i_2 = 4$ A (2) $i_1 = 8$ A, $i_2 = 4$ A
 (3) $i_1 = 4$ A, $i_2 = 4$ A (4) $i_1 = 3$ A, $i_2 = 12$ A
17. Six resistors each of 10 ohm are connected as shown. The equivalent resistance between points X and Y is:

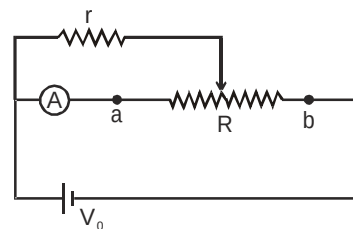


- (1) 20 ohm (2) 5 ohm
 (3) 25/3 ohm (4) 10 ohm
18. In the circuit, each resistance is 20Ω . The readings of A_1 , A_2 and A_3 are respectively :-

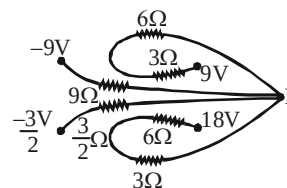


- (1) 3A, 1.8 A, 1.2 A (2) 3A, 1.2 A, 0.6 A
 (3) 3A, 0.6 A, 1.2 A (4) 3A, 0.6 A, 0.6 A

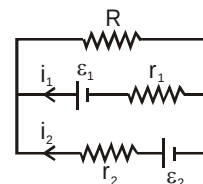
19. Referring to the circuit, R is the resistance of a potentiometer. As the sliding contact is moved from a to b, the reading in the ammeter will



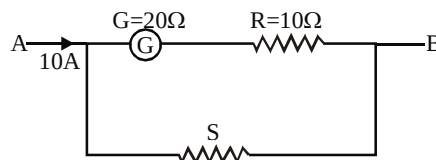
- (1) increase
 (2) decrease
 (3) initially decrease and then increase
 (4) initially increase and then decrease
20. The potential of point P in given diagram will be-



- (1) -2V (2) 2V (3) -1V (4) 1V
21. See the electrical circuit shown in this figure. Which of the following equations is a correct equation for it?

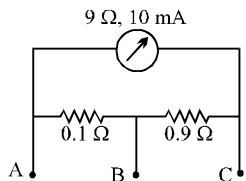


- (1) $\varepsilon_1 - (i_1 + i_2)R + i_1 r_1 = 0$
 (2) $\varepsilon_1 - (i_1 + i_2)R - i_1 r_1 = 0$
 (3) $\varepsilon_2 - i_2 r_2 - \varepsilon_1 - i_1 r_1 = 0$
 (4) $-\varepsilon_2 - (i_1 + i_2)R + i_2 r_2 = 0$
22. Full scale deflection current for galvanometer is 1 mA. What should be the value of shunt resistance (approximately) so that galvanometer shows half scale deflection for the circuit shown.

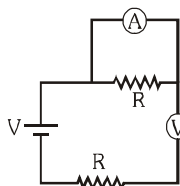


- (1) 1.5 m Ω (2) 3m Ω (3) 10 m Ω (4) 15 m Ω

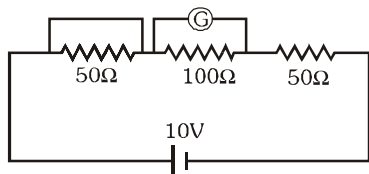
23. A milliammeter of range 10 mA and resistance $9\ \Omega$ is joined in a circuit as shown. The metre gives full-scale deflection for current I when A and B are used as its terminals, i.e., current enters at A and leaves at B (C is left isolated). The value of I is



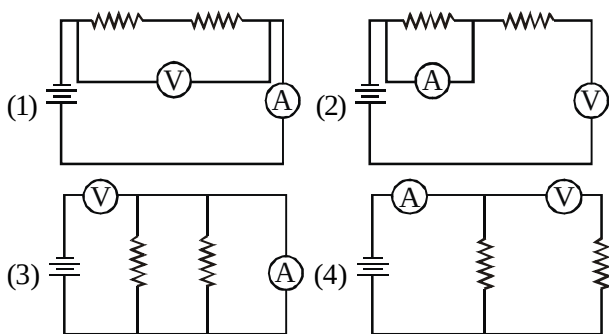
- (1) 100 mA (2) 900 mA (3) 1 A (4) 1.1 A
24. In the circuit shown, ammeter and voltmeter are ideal so reading of ammeter and voltmeter will be



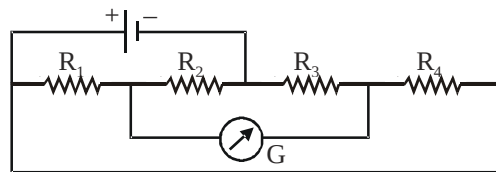
- (1) $\frac{V}{2R}, V$ (2) 0, V (3) 0, 0 (4) $\frac{V}{2R}, \frac{V}{2}$
25. In the circuit shown, resistance of galvanometer is $100\ \Omega$. It is being used as a voltmeter. Its reading will be



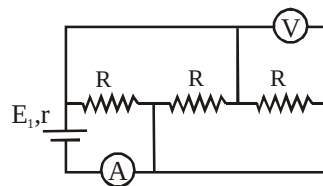
- (1) 0 volt (2) 2 volt (3) 5 volt (4) 10 volt
26. It is required to measure equivalent resistance of circuit with ideal battery, ideal voltmeter and ideal ammeter. Which circuit diagram shows voltmeter V and ammeter A correctly positioned to measure the total resistance of circuit.



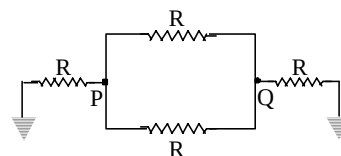
27. In the given circuit, the galvanometer G will show zero deflection if



- (1) $R_1 R_2 = R_3 R_4$ (2) $R_1 R_3 = R_2 R_4$
 (3) $R_1 R_4 = R_2 R_3$ (4) none of the above
28. In the circuit shown in figure, ammeter and voltmeter are ideal. If $E = 4V$, $R = 9\ \Omega$ and $r = 1\ \Omega$, then readings of ammeter and voltmeter are

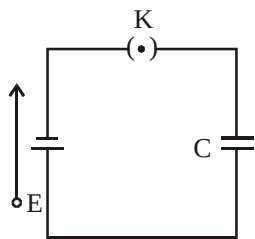


- (1) 1A, 3V (2) 2A, 3V
 (3) 3A, 4V (4) 4A, 4V
29. Out of the four bulbs of marking (i) 100 V, 40 W (ii) 200 V, 40 W (iii) 100 V, 60 W (iv) 200 V, 60 W, which bulb has highest resistance?
- (1) (i) (2) (ii)
 (3) (iii) (4) (iv)
30. The resistance between P and Q in the shown circuit is :



- (1) $\frac{R}{2}$ (2) $\frac{2R}{5}$
 (3) $\frac{3R}{5}$ (4) $\frac{R}{3}$
31. Find the quantity of heat evolved every second in a unit volume of a copper conductor with a current density of 30 A/cm^2 . ($\rho_{\text{copper}} = 2 \times 10^{-8}\ \Omega\text{m}$)
- (1) 1800 W/m³ (2) 600 W/m³
 (3) 300 W/m³ (4) 150 W/m³

32. A parallel plate capacitor is connected to a battery as shown in figure. Consider two situations:

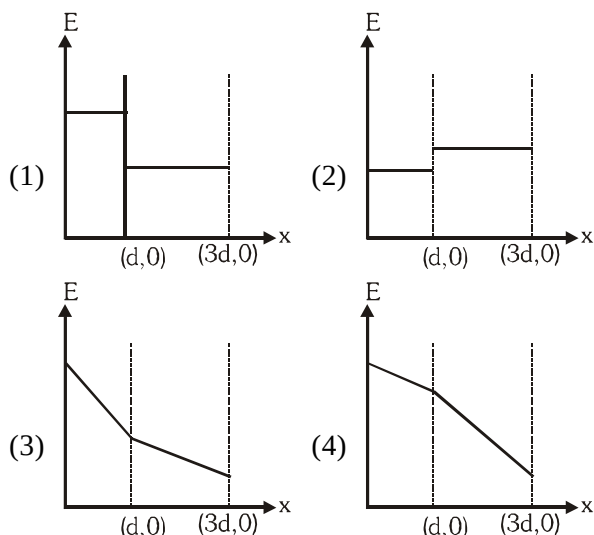
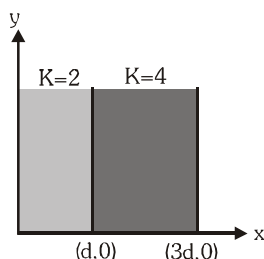


- A: Key K is kept closed and plates of capacitors are moved apart using insulating handle.
B: Key K is opened and plates of capacitors are moved apart using insulating handle.

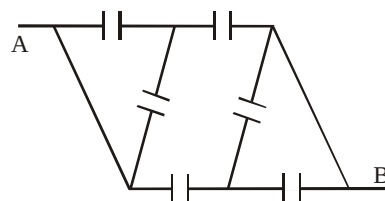
Choose the **CORRECT** option :-

- (1) In A : Q remains same but C changes.
(2) In B : V remains same but C changes.
(3) In A : V remains same and hence Q changes.
(4) In B : Q remains same and hence V remains constant.

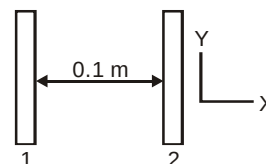
33. A parallel plate capacitor has two layers of dielectric as shown in figure. This capacitor is connected across a battery. The graph which shows the variation of electric field (E) & distance (x) from left plate.



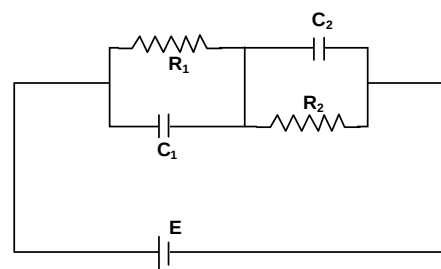
34. A network of six identical capacitors, each of value C , is made as shown in the figure. Equivalent capacitance between points A and B is



- (1) $C/4$ (2) $3C/4$ (3) $4C/3$ (4) $3C$
35. Two insulating plates are both uniformly charged in such a way that the potential difference between them is $V_2 - V_1 = 20$ V. (i.e., plate 2 is at a higher potential). The plates are separated by $d = 0.1$ m and can be treated as infinitely large. An electron is released from rest on the inner surface of plate 1. What is its speed when it hits plate 2 ?
($e = 1.6 \times 10^{-19}$ C, $m_0 = 9.11 \times 10^{-31}$ kg)

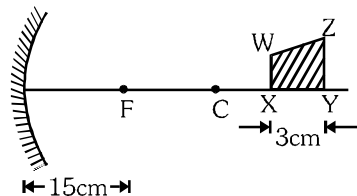


- (1) 2.65×10^6 m/s (2) 7.02×10^{12} m/s
(3) 1.87×10^6 m/s (4) 32×10^{-19} m/s
36. For the circuit shown in figure the ratio of energy stored in capacitor (1) to that of in capacitor (2) is



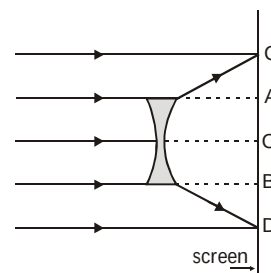
- (1) $\frac{R_1 C_1}{R_2 C_2}$ (2) $\frac{R_1 C_2}{R_2 C_1}$
(3) $\frac{R_1^2 C_1}{R_2^2 C_2}$ (4) $\frac{R_1 C_1^2}{R_2 C_2^2}$

37. To establish an instantaneous current of 2A through a $1\mu\text{F}$ capacitor; the potential difference across the capacitor plates should be changed at the rate of :
 (1) $2 \times 10^4 \text{ V/s}$ (2) $4 \times 10^6 \text{ V/s}$
 (3) $2 \times 10^6 \text{ V/s}$ (4) $4 \times 10^4 \text{ V/s}$
38. When an object is kept at a distance of 30 cm from a concave mirror, the image is formed at a distance of 10 cm. If the object is moved with a speed of 9 m/s, the speed with which image moves is
 (1) 10 cm/s (2) 1 m/s (3) 9m/s (4) 0.9m/s
39. Referring to the figure, the image of the quadrilateral XYZW placed in front of the concave mirror :



- (1) will be congruent of itself
 (2) will be similar to itself
 (3) will neither be congruent nor be similar
 (4) will be none of the above
40. Two identical thin planoconvex lenses of refractive index n are silvered, one on the plane side and the other on the convex side. The ratio of their focal lengths is :
 (1) $\frac{n}{n-1}$ (2) $\frac{n-1}{n}$ (3) $\frac{n+1}{n}$ (4) n
41. A hollow double concave lens is made of a very thin transparent material. It can be filled with water (refractive index μ_w) or either of two liquids L_1 or L_2 with refractive indices μ_1 and μ_2 respectively ($\mu_2 > \mu_w > \mu_1$). The lens will diverge a parallel beam of light incident on it, if it is filled with
 (a) L_2 and immersed in L_1
 (b) L_2 and immersed in water
 (c) water and immersed in L_1
 (d) air and immersed in either water or L_1 or L_2
 Select correct alternative :-
 (1) Only (a) and (b) (2) Only (a) and (d)
 (3) Only (a), (b) and (c) (4) Only (b), (c) and (d)

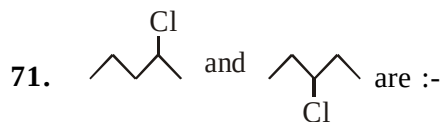
42. A concave lens is placed in the path of a uniform parallel beam of light falling on a screen as shown. Then,



- (1) intensity of light will be the same everywhere on the screen.
 (2) intensity in region AB will be smaller than what it would be in the absence of the lens
 (3) in the region AC and BD, the intensity will be smaller than what it would be in the absence of the lens
 (4) None of these
43. A plane wavefront AB is incident on a concave mirror as shown. Then, the wavefront just after reflection is
-
- (1) (2) (3) (4) None of the above
44. Two coherent sources emit light waves, which superimpose at a point where these can be expressed as
 $E_1 = E_0 \sin(\omega t + \pi/4)$
 $E_2 = 2E_0 \sin(\omega t - \pi/4)$
 here E_1 and E_2 are the electric field strengths of the two waves at the given point.
 If I is the intensity of wave expressed by field strength E_1 , find the resultant intensity.
 (1) I (2) $2I$ (3) $4I$ (4) $5I$
45. A screen is placed 50 cm from a single slit, which is illuminated with $6000 \text{ \AA}$ light. If distance between the first and third minima in the diffraction pattern is 3.00 mm, what is the width of the slit ?
 (1) 0.1 mm (2) 0.2 mm (3) 0.3 mm (4) 0.4 mm

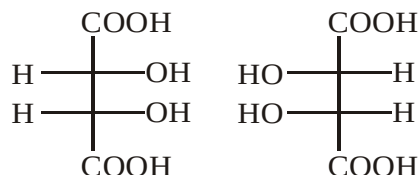
46. The order of ionisation potential between He^+ ion and H-atom (both species are in gaseous state) is:-
 (1) I. P. (He^+) = I. P. (H)
 (2) I. P. (He^+) < I. P. (H)
 (3) I. P. (He^+) > I. P. (H)
 (4) cannot be compared
47. Maximum number of ionisation energies possible for a carbon atom is :-
 (1) 1 (2) 2 (3) 4 (4) 6
48. Find out incorrect order of IP :-
 (1) $\text{O}^- < \text{S}^-$ (2) $\text{F}^- < \text{Cl}^-$
 (3) $\text{N}^- < \text{P}^-$ (4) $\text{Ne}^+ < \text{Ne}$
49. Incorrect set is :-
 (a) IE ($\text{Na} < \text{Li} < \text{B} < \text{Be}$)
 (b) EA, ($\text{N} < \text{P} < \text{O} < \text{S}$)
 (c) EN ($\text{S} < \text{O} < \text{Cl} < \text{F}$)
 (d) AR ($\text{F} < \text{N} < \text{O} < \text{Na}$)
 (1) a (2) a, b (3) a, b, c (4) c, d
50. Which of the following pairs of properties show reverse order of each other on moving along a period from left to right & from top to bottom in a group ?
 (1) Z_{eff} & E.A. (2) A.R. & E.A.
 (3) I.E. & E.A. (4) None
51. If the atomic number of an inert gas element is Z, then an element with which of the following atomic number will have the highest electronegativity according to Pauling scale ?
 (1) $Z - 2$ (2) $Z - 1$
 (3) $Z + 1$ (4) $Z + 2$
52. Intramolecular hydrogen bonding is not present in:-
 (1) Urea (2) Chloral hydrate
 (3) Salicylaldehyde (4) O-fluoro-phenol
53. Arrange the species according to increasing order of the no. of lone pair on their central atom from left to right :-
 (1) XeF_2 , ICl_3 , SO_2 (2) ICl_3 , SO_2 , XeF_2
 (3) SO_2 , ICl_3 , XeF_2 (4) SO_2 , XeF_2 , ICl_3
54. Which of the following has maximum $\text{X}-\hat{\text{P}}-\text{X}$ bond angle? (X = halogen atom)
 (1) PF_3 (2) PF_4^+
 (3) PCl_3 (4) POF_3
55. How many bond angles are of 90° in BrF_3 molecule?
 (1) 6 (2) 3 (3) 2 (4) 0
56. The value of dipole moment of BF_3 is similar to that of CH_4 . Hence structure of BF_3 will be:-
 (1) Trigonal planar (2) linear
 (3) Tetrahedral (4) Angular
57. The phosphate of a metal has the formula MHPO_4 . The formula of its chloride would be:-
 (1) MCl (2) MCl_2
 (3) MCl_3 (4) M_2Cl_2
58. In CsCl structure, 8 Cl^- ions surround one Cs^+ ion. Find out the total number of directional bond in CsCl .
 (1) 8 (2) 4 (3) 0 (4) 1
59. In which of the following d-orbitals are not used by central atom in hybridization ?
 (1) PF_5 (2) PCl_5
 (3) XeO_3 (4) XeF_6
60. Select correct statement about CH_4 .
 (1) All the hybrid orbitals of carbon are equivalent
 (2) All the hybrid orbitals of carbon have 25%, 2s character
 (3) All the hybrid orbitals of carbon are projected at the four vertex of a regular tetrahedron
 (4) All statements are correct
61. Match the following :-
- | Bonds | BE (kJ mole ⁻¹) |
|-----------|-----------------------------|
| (i) O-O | (p) 464 |
| (ii) O-H | (q) 138 |
| (iii) S-S | (r) 339 |
| (iv) S-H | (s) 213 |
- The correct match is
 (1) i-q, ii-p, iii-s, iv-r (2) i-q, ii-p, iii-r, iv-s
 (3) i-p, ii-q, iii-s, iv-r (4) i-r, ii-q, iii-p, iv-s
62. In which of the following processes, the value of magnetic moment does not change :-
 (1) $\text{N}_2 \rightarrow \text{N}_2^\ominus$ (2) $\text{N}_2 \rightarrow \text{N}_2^\oplus$
 (3) $\text{O}_2 \rightarrow \text{O}_2^+$ (4) $\text{O}_2^+ \rightarrow \text{O}_2^-$
63. Which of the following statement is incorrect :-
 (1) Bond order of CO > NO
 (2) N_2^+ is more stable than N_2^-
 (3) Bond order of O_2^+ is equal to N_2^+
 (4) F_2 is more stable than N_2

64. Correct order will be :-
 (1) $\text{LiH} > \text{NaH} > \text{KH} > \text{RbH}$ (Stability order)
 (2) $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3 > \text{BI}_3$ (Acidic order)
 (3) $\text{CaSO}_4 < \text{BaSO}_4 < \text{MgSO}_4$ (Solubility order)
 (4) $\text{Rb}^+ > \text{K}^+ > \text{Cs}^+ > \text{Na}^+$ (Ionic Mobility)
65. For the complex, $[\text{Fe}^{\text{II}}(\text{en})_3]_n [\text{Co}^{\text{III}}(\text{NH}_3)_2(\text{NO}_2)_4]_2$ the value of n is :-
 (1) 1 (2) 2 (3) 3 (4) 4
66. Which of the following will have three stereoisomeric forms ?
 (i) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ (ii) $[\text{CoCl}_3(\text{NH}_3)_3]$
 (iii) $[\text{CoCl}_2(\text{C}_2\text{O}_4)_2]$ (iv) $[\text{CoBrCl}(\text{en})_2]$
 (1) (iii) and (iv) (2) (i), (iii) and (iv)
 (3) (iv) only (4) All four
67. In compound $[\text{M}(\text{CO})_n]^z$, the correct match for highest M – C 'bond length' for given M, n and z respectively
- | M | n | z |
|--------|---|----|
| (1) Cr | 6 | 0 |
| (2) V | 6 | -1 |
| (3) Ti | 6 | -2 |
| (4) Mn | 6 | +1 |
68. In nitroprusside ion, iron and NO exist as Fe^{II} and NO^+ rather than Fe^{III} and NO. These forms can be differentiated by :-
 (1) estimating the concentration of iron
 (2) measuring the concentration of CN^-
 (3) measuring the solid state magnetic moment
 (4) thermally decomposing the compound
69. The formula for the compound tris (ethane-1, 2-diamine)cobalt (III) sulphate is
 (1) $[\text{Co}(\text{en})_3]\text{SO}_4$
 (2) $[\text{Co}(\text{SO})_4(\text{en})_3]$
 (3) $[\text{Co}(\text{en})_3](\text{SO}_4)_2$
 (4) $[\text{Co}(\text{en})_3]_2(\text{SO}_4)_3$
70. Which aqua complex do not show Jahn-Teller distortion?
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]^{+3}$ (2) $[\text{Cr}(\text{H}_2\text{O})_6]^{+2}$
 (3) $[\text{Cu}(\text{H}_2\text{O})_6]^{+2}$ (4) $[\text{Mn}(\text{H}_2\text{O})_6]^{+3}$



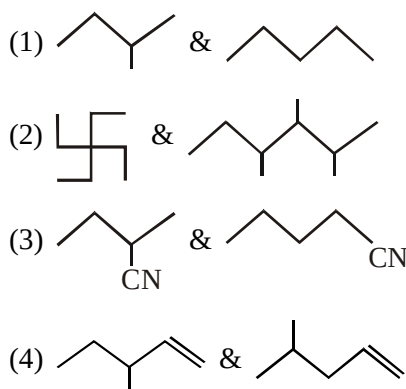
- (1) Position Isomers
 (2) Chain isomers
 (3) Functional Isomers
 (4) Metamers

72. The structure A and B represent :-

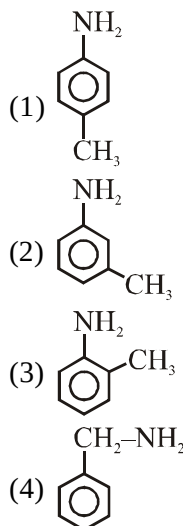


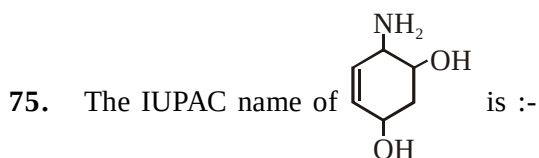
- (1) Enantiomer (2) Diastereomers
 (3) Homomers (4) Racemic Mixture

73. Which of the following is not a pair of chain isomers?



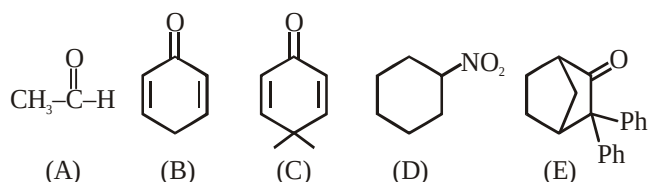
74. Which of the following is strongest base ?





- (1) 4-Aminocyclohex- 5-ene- 1, 3-diol
 (2) 4, 6-Dihydroxycyclohex- 2-enamine
 (3) 6-Aminocyclohex- 4-ene- 1, 3-diol
 (4) 2, 4-Dihydroxycyclohex- 2-enamine

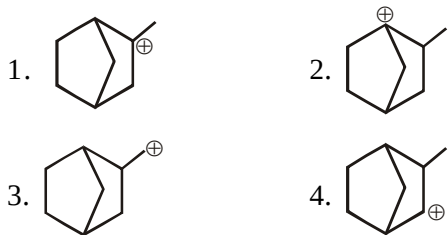
76. Consider following molecules :



Which can show tautomerism :

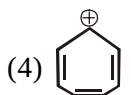
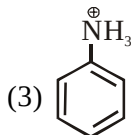
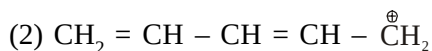
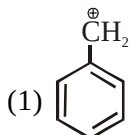
- (1) A, B, C (2) B, C, E
 (3) A, B, D (4) B, C, E, F

77. Find out correct stability order in the following carbocations :-



- (1) $4 > 1 > 3 > 2$ (2) $1 > 2 > 4 > 3$
 (3) $2 > 1 > 4 > 3$ (4) $1 > 4 > 3 > 2$

78. Which of the following positive charge not involved in resonance ?



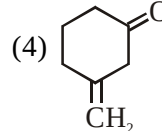
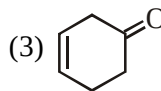
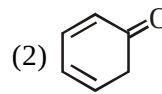
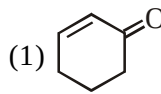
79. Sum of α -hydrogen in :-

- (i) 1-Hexene (ii) 2-Hexene (iii) 3-Hexene

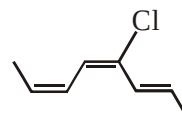
(i) + (ii) + (iii) = ?

- (1) 8 (2) 10 (3) 11 (4) 12

80. Which of the following has longest C-O bond?

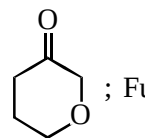
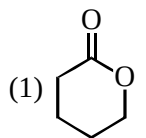


81. What is the correct IUPAC name of the following compound ?

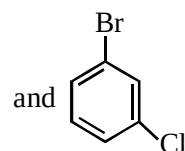
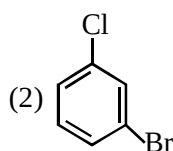


- (1) 6-chloro octa-2,4,6-triene
 (2) 5-chloro octa-2,4,6-triene
 (3) 4-chloro octa-2,4,6-triene
 (4) 3-chloro octa-2,4,6-triene

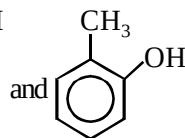
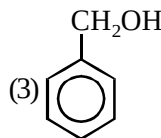
82. Which of the following relation is incorrect:



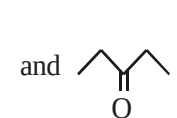
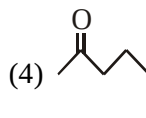
; Functional group isomer



; Identical

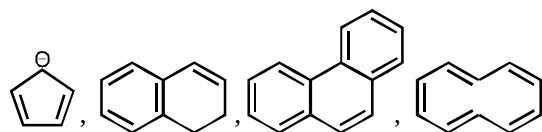
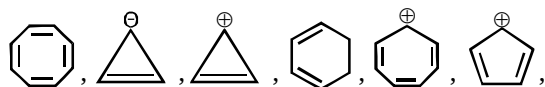


; Functional group isomer



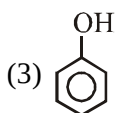
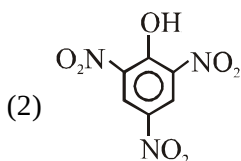
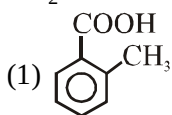
; Chain isomer

83. Amongst the following, the number of aromatic compound (s) are :



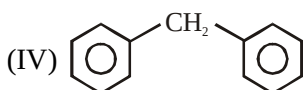
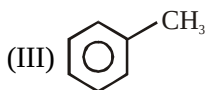
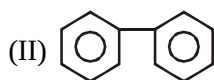
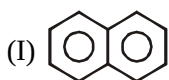
- (1) 5 (2) 6 (3) 4 (4) 3

84. Which of the following gives effervescence of CO_2 on reaction with NaHCO_3



- (4) Both (1) & (2)

85. Which of the following have only 2°H atom :-



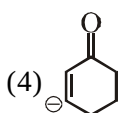
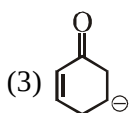
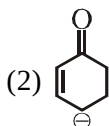
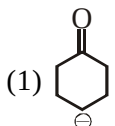
Correct code is :

- (1) I & II (2) I, II, IV
(3) I, III, IV (4) I, II, III, IV

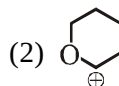
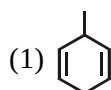
86. Which of the molecules has two sigma (σ) and two pi (π) bonds :

- (1) C_2H_4 (2) $\text{C}_2\text{H}_2\text{Cl}_2$ (3) HCN (4) C_3H_4

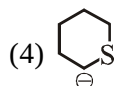
87. Most stable carbanion is :



88. Resonance is not possible in :



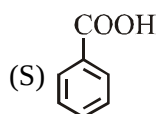
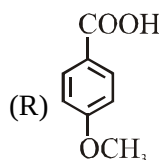
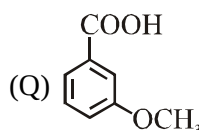
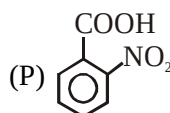
- (3) $\text{CH}_2=\text{CH}-\text{Cl}$



89. The hybridisation of nitrogen in (pyrrole) is:

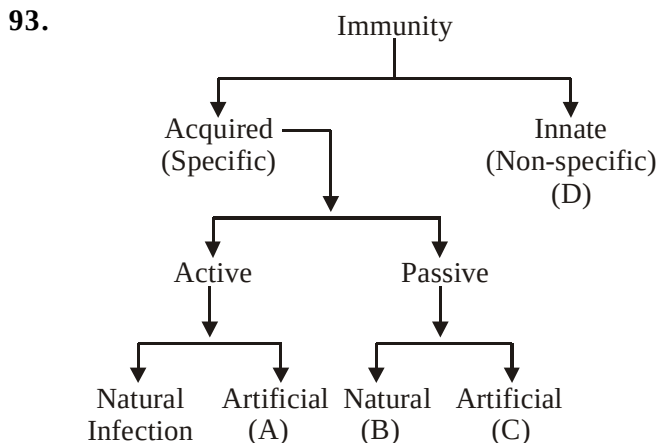
- (1) sp^3 (2) sp^2
(3) sp (4) Can't be predicted

90. The decreasing order of acidity of following benzoic acid derivatives is :



- (1) $\text{S} > \text{Q} > \text{R} > \text{P}$
(2) $\text{R} > \text{Q} > \text{P} > \text{S}$
(3) $\text{Q} > \text{R} > \text{S} > \text{P}$
(4) $\text{P} > \text{Q} > \text{S} > \text{R}$

- 91.** Study of antigen - antibody reaction is called -
 (1) Immunology (2) Serology
 (3) Allergy (4) Both (1) and (3)
- 92.** How we define health?
 (1) Absence of disease only
 (2) Physical fitness only
 (3) Mental fitness only
 (4) State of complete physical, mental and social well - being



Which labelling is related with Antiserum -

- (1) A (2) B
 (3) C (4) D
- 94.** Which prevents entry of microbes in body ?
 (1) Skin (2) Mucosa
 (3) pH of Stomach (4) Fever
- 95.** Incorrect about interferons is -
 (1) Cellular barrier
 (2) Cytokine barrier
 (3) species specific
 (4) not virus specific
- 96.** Army of proteins in response to pathogens into our blood to fight with them are secreted by -
 (1) T_H - cells
 (2) T_k - cells
 (3) B - lymphocytes
 (4) N.K cells
- 97.** Heaviest antibody is ?
 (1) IgG (2) IgM
 (3) IgA (4) IgD

- 98.** Read the following statements :-
 (A) Passive immunity serves no purpose in immunodeficient host.
 (B) Active immunity is used for the treatment of infections.
 (C) Active immunity develops after a lag period.
 (D) Passive immunity is conferred by introduction of antigen in the body.
 How many of the above statements are false ?
 (1) 3 (2) 4 (3) 2 (4) 1
- 99.** A female suffering from patau syndrome will have how many barr bodies ?
 (1) 2 (2) 1 (3) 0 (4) 3
- 100.** CABG (vascular grafting) is an example of
 (1) Isografting (2) Autografting
 (3) Homografting (4) Xenografting
- 101.** Mark the incorrect matching :-
 (1) Down syndrome - Aneuploidy
 (2) Turner syndrome - 45 with XO
 (3) Cri-du-chat - Trisomy in chromosome 5th
 (4) Klinefelter syndrome - 44 + XXY
- 102.** Find odd one out for gene mutation -
 (1) PKU
 (2) Albinism
 (3) Alkaptonurea
 (4) G - 6 - P - D deficiency syndrome
- 103.** Pannus Granules are formed in ?
 (1) Masthenia gravis
 (2) IDDM
 (3) Rheumatoid arthritis
 (4) Pernicious anemia
- 104.** Salk polio vaccine is _____
 (1) Live (2) Killed
 (3) Toxoid (4) Recombinant
- 105.** Primary lymphoid organ is -
 (1) Bone marrow (2) Spleen
 (3) Tonsils (4) MALT
- 106.** Which drug has potent stimulating action on central nervous system ?
 (1) Barbiturate (2) Benzodiazepines
 (3) Cocaine (4) Heroin

107. Which statement is false ?

- (1) Dependence is the tendency of the body to manifest a characteristic and unpleasant withdrawal syndrome if regular doses of drugs/alcohol is abruptly discontinued.
- (2) Because of the perceived benefits, drugs are frequently used repeatedly.
- (3) Use of drugs even once, always leads to coma and death due to respiratory failure, heart failure and cerebral haemorrhage.
- (4) Smoking also paves way to hard drugs.

108. Intestinal perforation and death may occur in severe cases of which disease ?

- (1) Pneumonia (2) Typhoid
- (3) Filariasis (4) Malaria

109. Malignant malaria is caused by ?

- (1) *Plasmodium vivax*
- (2) *Plasmodium falciparum*
- (3) *Plasmodium malariae*
- (4) Retro virus

110. In the given list how many diseases are spread by female mosquito vectors ?

Malaria, dengue, chikungunya, elephantiasis, ringworms

- (1) 4 (2) 3 (3) 2 (4) 5

111. "Infected cell can survive while viruses are being replicated and released". In case of HIV infection given statement is true for which cell ?

- (1) Macrophage (2) T-helper
- (3) T-killer (4) B-lymphocyte

112. Which pair is not matched correctly ?

- (1) Antihistaminic drug $\Rightarrow$ cetirizine
- (2) Opioid analgesic $\Rightarrow$ Diclofenac
- (3) Antipyretic drug $\Rightarrow$ Paracetamol
- (4) Antibiotic $\Rightarrow$ Chloramphenicol

113. Which drug inhibits the integrase enzyme of the HIV?

- (1) Ritonavir (2) Zidovudin
- (3) Stavudin (4) Elvitegravir

114. Which disease is not caused by a bacteria ?

- (1) Gonorrhoea (2) Syphilis
- (3) Trichomoniasis (4) Chlamydia

115. Which pair is not matched correctly ?

- (1) Tetanus $\Rightarrow$ Opisthotonus
- (2) Diphtheria $\Rightarrow$ Schick test
- (3) Hepatitis-B virus $\Rightarrow$ RNA virus
- (4) Typhoid $\Rightarrow$ Widal test

116. Which disease is completely curable if detected early and treated properly ?

- (1) Chlamydia (2) Genital herpes
- (3) AIDS (4) Hepatitis-B

117. Which structure of HIV has complementary sequence to CD-4 receptors ?

- (1) GP-41 (2) P-24 (3) P-17 (4) GP-120

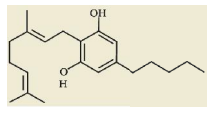
118. Burkitt's lymphoma is caused due to reciprocal translocation between which chromosomes :-

- (1) 22 and 9 (2) 8 and 14
- (3) 8 and 22 (4) 14 and 9

119. Which is the most feared property of cancer ?

- (1) Contact inhibition
- (2) non-capsulated
- (3) Metastasis
- (4) inhibition of telomerase activity

120. Which pair is matched correctly ?

(1)		Flowering branch of Datura
(2)		Chemical structure of morphine
(3)		Leaves of Cannabis sativa
(4)		Opium poppy

121. Which of the following trait of pea plant can express itself only in homozygous state ?

- (1) Tall plant (2) Axial flower
- (3) White flower (4) Green pod

- 122.** When red flowered plants of snapdragon are crossed with pink flowered plants then ratio of progenies obtained will be :-
 (1) All pink (2) 1 red : 1 pink
 (3) 1 red : 1 white (4) 1 red : 2 pink : 1 white
- 123.** What can be the blood group of offsprings when both the parents have AB blood group?
 (1) A and B only (2) AB only
 (3) A, B, AB and O (4) A, B and AB
- 124.** How many type of gametes will be produced by a diploid organism which is heterozygous for 4 loci?
 (1) 4 (2) 8 (3) 16 (4) 32
- 125.** In hybridisation experiment, 27 genotypes give _____ F_2 phenotypes :-
 (1) Four (2) Eight
 (3) Sixteen (4) Two
- 126.** In F_2 generation of dihybrid cross of mendel, the ratio between recombinants is :-
 (1) 9 : 3 : 3 : 1 (2) 3 : 1
 (3) 2 : 1 (4) 1 : 1
- 127.** The phenotypic ratio in F_2 generation for flower colour of sweet pea is :-
 (1) 9 : 3 : 3 : 1 (2) 9 : 3 : 4
 (3) 9 : 7 (4) 12 : 3 : 1
- 128.** The punnett square shown below represents the pattern of inheritance in dihybrid cross when yellow (Y) is dominant over green (y) and round (R) is dominant over wrinkled (r) seeds :-

	RY	Ry	rY	ry
RY	P	Q	R	S
Ry	T	U	V	W
rY	X	Y	Z	K
ry	L	M	N	O

Find out the odd one :-

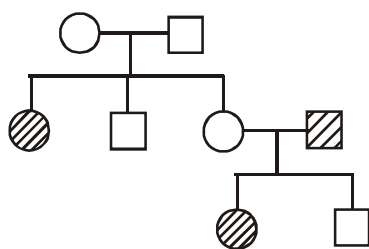
- (1) S (2) Y (3) N (4) L
- 129.** A plant can produce red flowers only when both gene A and B are present in its genotype. Presence of only A or only B result in blue flower. Complete absence of dominant alleles result in white flower. A plant having genotype AaBb was self pollinated. What percentage of progeny will produce red flowers?
 (1) 18% (2) 23.5% (3) 56.25% (4) 37.50%

- 130.** Read the following statements (A-D) :-
 (A) All gene mutation result in non-synthesis of normal product.
 (B) Polygenic inheritance also takes into account the influence of environment
 (C) Genes present on different loci always control different characters.
 (D) In case of back cross, genotype ratio and phenotype ratio are always same.
 How many of the above statements are wrong?
 (1) Four (2) Three
 (3) Two (4) One
- 131.** Semi-conservative mode of DNA replication in eukaryotes was proved by :-
 (1) Taylor (2) Meselson and Stahl
 (3) Watson (4) Crick
- 132.** In 1900 A. D. three biologists independently rediscovered Mendel's principles. They are :-
 (1) De Vries, Correns and Tschermak
 (2) Sutton, Morgan and Bridges
 (3) Avery, McLeod and McCarty
 (4) Bateson, Punnett and Bridges
- 133.** Match the following (column-I with Column-II)

Column-I		Column-II	
a	Allele	i	Two alleles of a gene are identical
b	Genotype	ii	An individual having two different alleles of a gene
c	Homozygous	iii	Genetic constitution of an organism
d	Heterozygous	iv	Alternative forms of a gene

- (1) a-i, b-ii, c-iii, d-iv
 (2) a-iv, b-iii, c-i, d-ii
 (3) a-iii, b-iv, c-ii, d-i
 (4) a-iii, b-ii, c-iv, d-i

134. Given pedigree represent :-



- (1) Autosomal recessive
- (2) Autosomal dominant
- (3) x linked dominant
- (4) y linked

135. Mendelian principles are :-

- (1) Linkage, segregation and independent assortment
- (2) Dominance, segregation and linkage
- (3) Dominance, segregation and independent assortment
- (4) Dominance, independent assortment and linkage

136. Gene A and B are linked and 24 cM apart on chromosome. An offspring of parents AAbb and aaBB will produces what percent of various gametes?

- (1) 38% AB, 12% Ab, 12% aB, 38% ab.
- (2) 12% AB, 38% Ab, 38% aB, 12% ab.
- (3) 24% AB, 26% Ab, 26% aB, 24% ab.
- (4) 26% AB, 24% Ab, 24% aB, 26% ab.

137. Henking observed that in few insects (Grass hopper) male, 50% of the sperm received a specific nuclear structure known as X - body, whereas other 50% sperms did not receive it. This method of sex determination is :-

- (1) XX - XY method
- (2) XX - XO method
- (3) ZZ - ZW method
- (4) Haplodiploidy

138. Both husband and wife have normal vision, though their fathers were colourblind. The probability of their daughter becoming colourblind will be :-

- (1) 0%
- (2) 25%
- (3) 50%
- (4) 75%

139. In polygenic inheritance :-

- (1) $\frac{n(n+1)}{2}$ formula can be used to find out total possible genotypes
- (2) $(2n+1)$ formula can be used to find out total possible genotypes.
- (3) Number of intermediates increases with increase in number of genes.
- (4) To find out contribution of each dominant allele, maximum expression of character must be known.

140. In mendels dihybrid cross in F_2 generation what proportion of progeny show parental genotype ?

- (1) 9/16
- (2) 3/16
- (3) 4/16
- (4) 2/16

141. Which of the following is not an essential amino-acid?

- (1) Threonine
- (2) Lysine
- (3) Methionine
- (4) Glutamic acid

142. DNA-m-RNA hybrid are made by using :-

- (1) Reverse transcriptase
- (2) DNA-polymerase
- (3) Restriction endonuclease
- (4) DNA-ligase

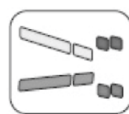
143. According to Hardy-Weinberg law which one is incorrect ?

- (1) Applicable for large population
- (2) Absence of Mutation
- (3) Random mating
- (4) $p^2 \times q^2 \times 2pq = 1$

144. Which of the following statement is correct ?

- (1) hn RNA is subjected to a process called splicing where the exons are removed.
- (2) In capping an unusal nucleotides is added to the 3' end of hn RNA
- (3) Splicing represents the dominance of DNA world
- (4) In tailing, adenylated residues are added at 3' end of hnRNA in a template independent manner.

- 145.** For the genetic material which criteria is not essential for a biomolecule :-
 (1) It should be able to generate its replica
 (2) It should have catalytic property
 (3) It should be chemically and structurally stable
 (4) It should be able to express itself in the form of mendelian character
- 146.** Several proteins may be produced at the same time from a single m-RNA in prokaryotes by :-
 (1) the action of several ribosomes on a string called polyribosome
 (2) several DNA polymerase molecules working sequentially
 (3) single peptides that associate ribosomes with rough E.R.
 (4) the involvement of multiple spliceosomes complex
- 147.** Minimum number of genes present on which chromosome number in human ?
 (1) 1st chromosome (2) Y-chromosome
 (3) X-chromosome (4) 9th chromosome
- 148.** Which of the following RNA has structural and catalytic role in translation ?
 (1) m-RNA (2) t-RNA
 (3) r-RNA (4) g-RNA
- 149.** Which is not found in DNA ?
 (1) Deoxy ribose sugar (2) 5-methyl uracil
 (3) Demethylated thymine (4) H₃PO₄
- 150.** A segment of DNA molecule contains 20,000 base pairs. What is the number of phosphate molecule in this DNA molecule ?
 (1) 20,000 (2) 40,000 (3) 10,000 (4) 5,000
- 151.** Severo ochoa enzyme is :-
 (1) Polyribonucleotide phosphorylase
 (2) Polyribonucleotide dephosphorylase
 (3) Polydeoxyribo nucleotide dephosphorylase
 (4) Polydeoxyribo nucleotide phosphorylase
- 152.** Peptide bond formation is catalysed by :-
 (1) 18s rRNA (2) 5s rRNA
 (3) 5.8s rRNA (4) 28s rRNA
- 153.** Stop codon are recognised by :-
 (1) tRNA (2) Release factor
 (3) UTR (4) Amino acid
- 154.** An opposite phenomenon of linkage is :-
 (1) Dominance (2) Segregation
 (3) Co-dominance (4) Crossing over
- 155.** Real Inducer of the lac operon is :-
 (1) Lactose (2) Allolactose
 (3) Glucose (4) Galactose
- 156.** On an average human gene consists of :-
 (1) 3000 bases (2) 1.4 million bases
 (3) 2.4 million bases (4) 29 bases
- 157.** Which of the following is not used in the DNA finger printing :-
 (1) Gel electrophoresis (2) Northern blotting
 (3) PCR (4) Radioactive probe
- 158.** Find out the incorrect for AUG :-
 (1) Dual function
 (2) Initiator codon
 (3) Recognised by RF (Releasing factor)
 (4) Code for methionine
- 159.** Match the following :-
- | | Column I | | Column II |
|---|---------------------|-----|-------------------------|
| a | Sickle cell anaemia | i | Deletion |
| b | Thalassemia | ii | Translocation |
| c | CML | iii | Lethal genetic disorder |
| d | Cry du chat | iv | Substitution |
- (1) a-iv, b-iii, c-ii, d-i (2) a-iii, b-iv, c-ii, d-i
 (3) a-ii, b-iii, c-i, d-iv (4) a-iii, b-iv, c-i, d-ii
- 160.** Triplet nature of genetic code were suggested by :-
 (1) George Gamow (2) H.G. Khorana
 (3) Nirenberg (4) F.Crick
- 161.** Observe the diagrams given below and identify the wrongly matched option.

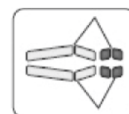


A

- (1) A-G₁ phase
 (3) C-Meiosis-II

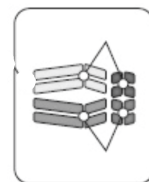


B



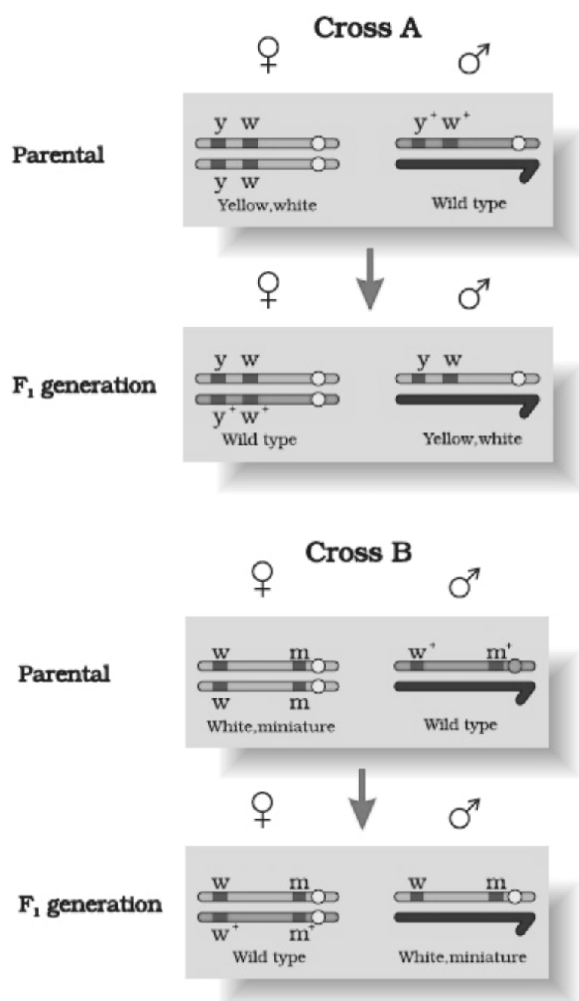
C

- (2) D-G₂ phase
 (4) B-Gamete (germ cell)



D

162. Observe the given crosses (A and B) done in *Drosophila*.



Select the incorrect statement.

- (1) y and w are more linked than w and m
 - (2) In cross B, parental type will be more than recombinants
 - (3) Recombinants will be more in number in cross A than cross B
 - (4) In cross A, parental type will be more than the recombinants
163. Select the set of wrong statements:
- A–Coplanarity of stacks of nucleobases, confers additional stability to DNA double helix.
- B–Mendel succeeded as he has taken small sampling size for experiments.

C–Mendelian factors are actually chromosomes.

D–ABO Blood group is an example of multiple allelism.

- (1) A, B, C
- (2) B, C, D
- (3) A, C, D
- (4) B, C

164. Observe the phenotypic effects of a gene with 2 alleles–B and b in pea plant.

Genotype	Grain Size	Grain Shape
BB	Large	Round
Bb	Intermediate	Round
bb	Small	Wrinkled

Which of the following extract is incorrect in concern to above facts?

- (1) Gene shows co-dominance
 - (2) Gene shows normal dominant recessive relationship for grain shape
 - (3) Gene is example of pleiotropic gene
 - (4) Gene shows incomplete dominance for grain size
165. Life cycle of fruitfly is:
- (1) 1 week
 - (2) 2 weeks
 - (3) 1 month
 - (4) 2 months

166. Match the columns:

Column A		Column B	
A	Taylor and colleagues	(i)	Biochemical characterization of transforming principle
B	Meischer	(ii)	Nuclein
C	Crick	(iii)	Central dogma
D	Avery, Macleod and McCarthy	(iv)	Semi-conservative replication of DNA

- (1) A–(iv), B–(ii), C–(i), D–(iii)
 - (2) A–(iv), B–(ii), C–(iii), D–(i)
 - (3) A–(ii), B–(iv), C–(i), D–(iii)
 - (4) A–(iv), B–(i), C–(iii), D–(ii)
167. Discontinuous fragments of DNA which are formed during replication of DNA are joined by:
- (1) Ligase
 - (2) Lyase
 - (3) DNA polymerase
 - (4) RNA polymerase

168. Match the items in Column-I with those in Column-II:

Column-I		Column-II	
A	r-RNA	(i)	Termination factor
B	m-RNA	(ii)	Catalytic role in translation
C	P	(iii)	Reads the genetic code
D	t-RNA	(iv)	Provides the template
E	σ	(v)	Initiation factor

- (1) A-(iv), B-(iii), C-(v), D-(ii), E-(i)
 (2) A-(iv), B-(ii), C-(i), D-(iii), E-(v)
 (3) A-(iii), B-(ii), C-(v), D-(iv), E-(i)
 (4) A-(ii), B-(iv), C-(i), D-(iii), E-(v)

169. Except some codons in1.... (an organelle) and in some2.... (group of organisms). all other codons are almost universal.

- (1) Chloroplast, Bacteria
 (2) Mitochondria, Protozoans
 (3) ER, Fungi
 (4) GB, Slime moulds

170. Read the following steps involved in translation and arrange them in correct sequence.

- (A) Chain elongation (B) Initiation
 (C) Charging of t-RNA (D) Termination
 (E) Activation of amino acids
 (F) Peptide bond formation

- (1) B $\rightarrow$ E $\rightarrow$ A $\rightarrow$ C $\rightarrow$ D $\rightarrow$ F
 (2) B $\rightarrow$ E $\rightarrow$ C $\rightarrow$ F $\rightarrow$ A $\rightarrow$ D
 (3) B $\rightarrow$ E $\rightarrow$ F $\rightarrow$ C $\rightarrow$ A $\rightarrow$ D
 (4) E $\rightarrow$ C $\rightarrow$ A $\rightarrow$ F $\rightarrow$ B $\rightarrow$ D

171. Select the incorrect statement :

- (1) Operons are common in prokaryotes
 (2) Operons may possess few structural genes
 (3) 3 regulatory genes are found in lac operon
 (4) Regulation of lac operon is negative regulation

172. Select the incorrect statement :

- (1) 22nd chromosome was last to be sequenced in human
 (2) There are about 20,000-25,00 genes in human
 (3) Some levels of 'lac operon' is always expressed in E.coli
 (4) Human genome project is closely associated with bioinformatics

173. Match the columns:

Column A		Column B	
1	Sanger	A	DNA finger printing
2	Alec Jaffereys	B	Operon
3	Jacob and Monad	C	Automated DNA sequences
4	Crick	D	Central dogma

- (1) 1-C, 2-A, 3-B, 4-D (2) 1-C, 2-A, 3-D, 4-B
 (3) 1-A, 2-C, 3-D, 4-B (4) 1-A, 2-B, 3-D, 4-C

174. Backbone of DNA is formed by:

- (1) Phosphodiester bonds (2) Glycosidic bonds
 (3) Amide bonds (4) H-bonds

175. All of the following are polymeric secondary metabolites except:

- (1) Cellulose (2) Gums
 (3) Rubber (4) Codeine

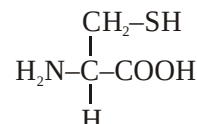
176. Which of the following is not correctly matched with its category?

- (1) Lecithin-Glycoprotein
 (2) GLUT-4-Protein
 (3) Chitin-Carbohydrates
 (4) Adenylic acid-Nucleotide

177. Identify the mismatch w.r.t. the level of organization in structure of protein (Column I) with its feature / type of folding (Column II) and examples (Column III) :

	Column I	Column II	Column III
(1)	Primary structure	Linear sequence of amino acids	Polypeptide on Ribosome
(2)	Secondary structure	Helix or pleated form	Fibrous protein
(3)	Quaternary structure	2 or more polypeptide	Fibroin
(4)	Tertiary structure	Hollow woolen ball	Most of Enzyme

178. Following structure is related to which amino acid?



- (1) Arginine (2) Tryptophan (3) Alanine (4) Cysteine

179. Which of the following content is present in acid soluble pool ?

- (1) RNA (2) Cellulose
 (3) Glucose (4) Protein

180. Which of the following is microbiomolecule still found in a macromolecular fraction ?

- (1) Polysaccharide (2) Protein
 (3) Nucleic acid (4) Lipid

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3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
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